



District-Level Market Attractiveness Index (MAI)

A Machine Learning Framework for Optimizing Pharmaceutical Resource Allocation
Strategic Expansion and Territory Planning in India
Prepared for Sun Pharma Evaluation Panel

Executive Summary

- The Challenge: India's immense socio-economic diversity makes relying on historical sales insufficient for growth.
- The Solution: An algorithmic Market Attractiveness Index (MAI) utilizing a robust Machine Learning data pipeline.
- The Output: Three distinct indices (Overall, Chronic, Acute) guiding highly targeted, therapy-specific deployment.

Business Objectives & Strategic Imperatives

- Precision Resource Allocation: Optimize Medical Representative (MR) deployment based on mathematical viability.
- Segmented Therapy Strategy: Decouple sales strategies for Acute (volume) and Chronic (value) drug portfolios.
- Predictive Opportunity: Identify 'hidden gem' districts with rising economic and epidemiological trends.

Market Attractiveness Index (MAI) – Core Definition

- A composite, standardized score (0 to 100) quantifying the commercial potential of a district.
- Normalizes disparate metrics (GDP vs. Disease Prevalence) into a unified business score.
- Drives investment prioritization: Score 95 = Tier-1 priority; Score 35 = Lean distribution approach.

1. Overall Pharmaceutical MAI

- Represents a holistic, blended view of a district's total healthcare capacity and wealth.
- Captures both short-term volume potential and long-term value potential.
- Used for broad footprint planning and determining regional distribution hubs (C&F agents).

2. Chronic Therapy MAI

- Targets commercial viability for daily, recurring medications (cardiovascular, anti-diabetic).
- Heavily dependent on Purchasing Power (GDP), Specialized Infrastructure, and Lifestyle Disease indicators.
- Guides deployment of highly specialized senior MRs to wealthy, urbanized districts.

3. Acute Therapy MAI

- Maximizes volume, reach, and immediate care for short-term medications (antibiotics, pain management).
- Relies on total population mass, younger demographics, and basic healthcare facility volume.
- Guides mass-market distribution strategies focusing on pharmacies and general physicians.

Variable Selection: Economic Indicators

- Average GDP & Per Capita Wealth: Direct indicators of out-of-pocket healthcare purchasing power.
- Literacy Rates: Correlates strongly with health awareness and proactive medical seeking behavior.
- Rationale: Wealthier districts can sustain high-margin chronic portfolios.

Variable Selection: Infrastructure Readiness

- Total Healthcare Facilities: Measures the physical capacity for diagnosis and prescription.
- Public vs. Private Ratios: Indicates the level of premium healthcare access.
- Rationale: You cannot sell specialized drugs if the district lacks the clinics to prescribe them.

Variable Selection: Epidemiological Data

- Blood Sugar & Blood Pressure Levels: Directly maps to anti-diabetic and cardiovascular portfolios.
- Tobacco Use & Age Demographics: Predicts respiratory and generalized acute illness patterns.
- Rationale: Replaces 'gut feeling' with verified public health statistics.

Contrasting Drivers: Acute vs. Chronic

- Chronic Drivers: High GDP, High Literacy, High Hypertension rates, Aging population.
- Acute Drivers: Total Population density, High clinic counts, Younger demographics.
- Business Impact: Prevents misallocating expensive resources into low-yield territories.

Methodology: Data Pipeline & Preparation

- Data Sourcing: Aggregated from verified government health and economic repositories.
- Data Engineering: Merged distinct regional datasets using Python (Pandas).
- Transformation: Engineered new features to represent per-capita ratios rather than raw counts.

Methodology: Handling Missing Data

- Challenge: Real-world government data contains gaps and null values.
- Solution: Implemented SimpleImputer using Median Imputation.
- Why Median?: Prevents statistical skewing caused by extreme outliers in wealth or population.

Methodology: Normalization Strategy

- The Issue: Comparing GDP (billions) with disease prevalence (percentages) creates scale imbalances.
- The Fix: Applied StandardScaler (Z-score normalization).
- The Result: Every variable is converted to a standard scale, allowing the algorithm to weight them fairly.

Methodology: Index Construction (PCA)

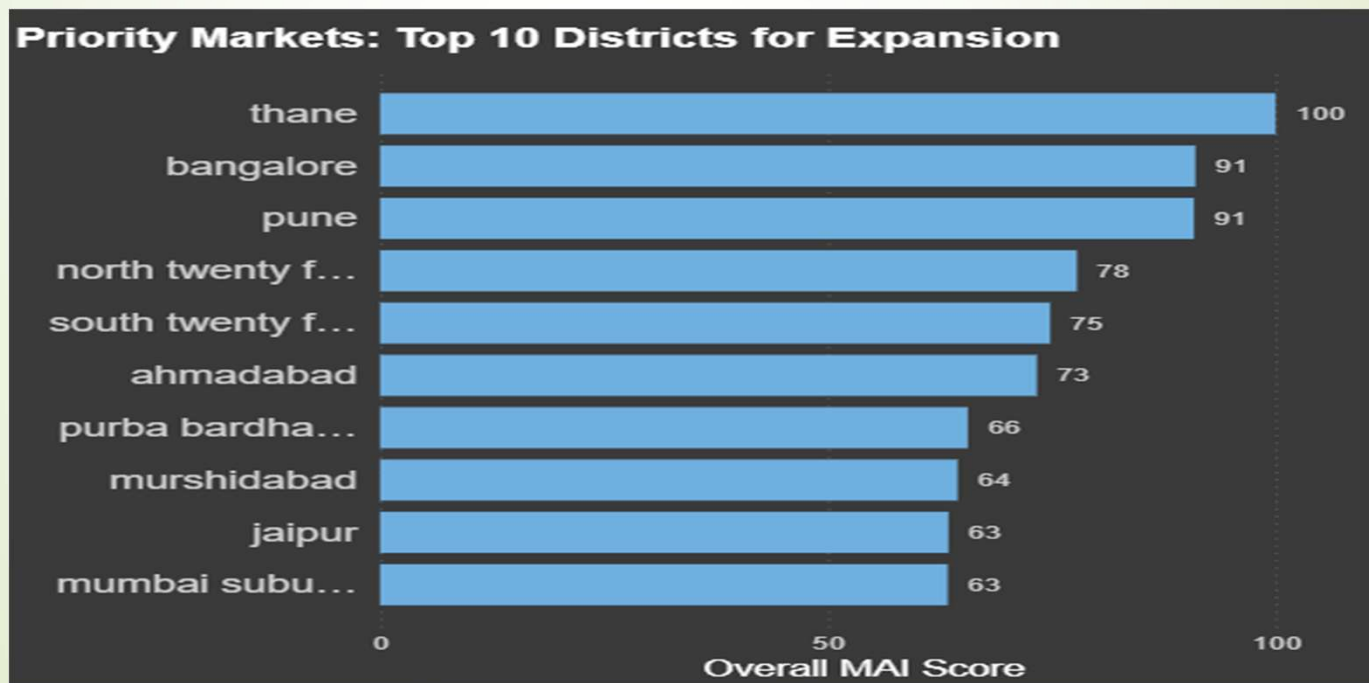
- Algorithm Used: Principal Component Analysis (PCA).
- Process: Extracted the 1st Principal Component to capture maximum variance across all variables.
- Final Output: Used MinMaxScaler to convert the complex PCA outputs into a clean 0-100 business score.

Methodology: Future Demand Modeling

- Population Scaling: Factored in birth rates and age bracket shifts.
- Economic Trajectory: Analyzed wealth growth to predict future purchasing power.
- Predictive Output: Highlights which districts will age into chronic disease hubs over the next 5 years.

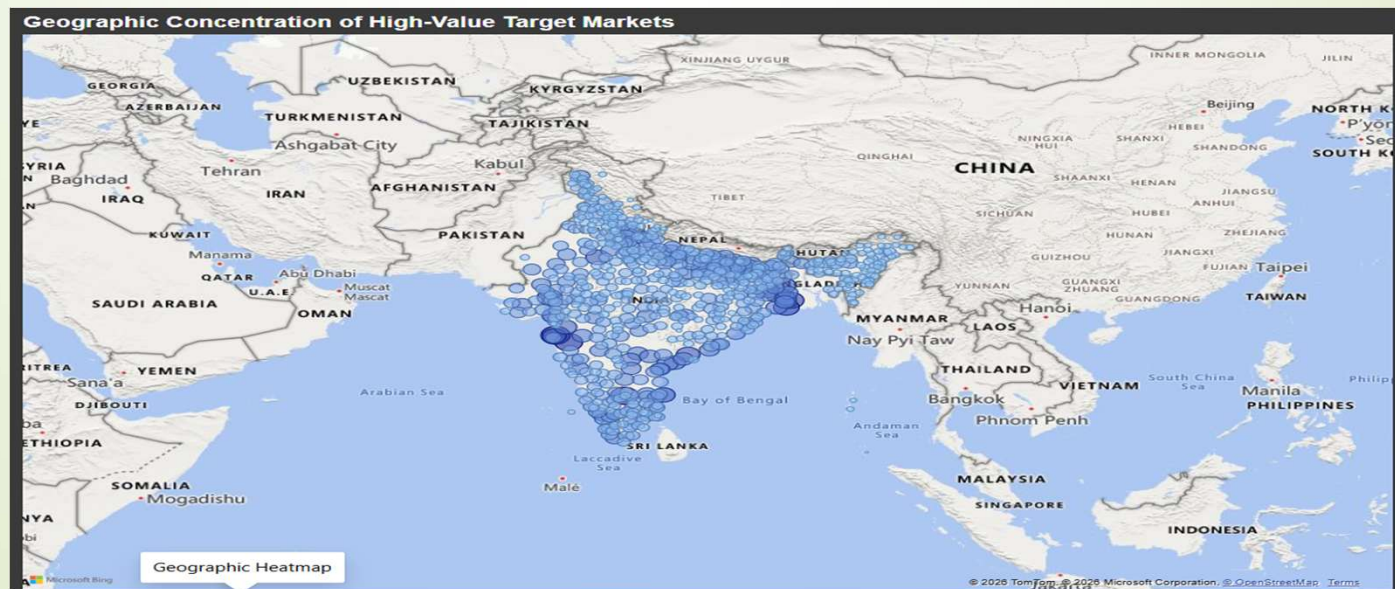
Visual Insights: District Rankings

- Format: Top 10 Clustered Bar Chart for immediate priority targeting.
- Executive Value: Provides a definitive, rank-ordered 'Hit List' for regional sales directors.
- Action: Immediate deployment of senior resources to top decile districts.



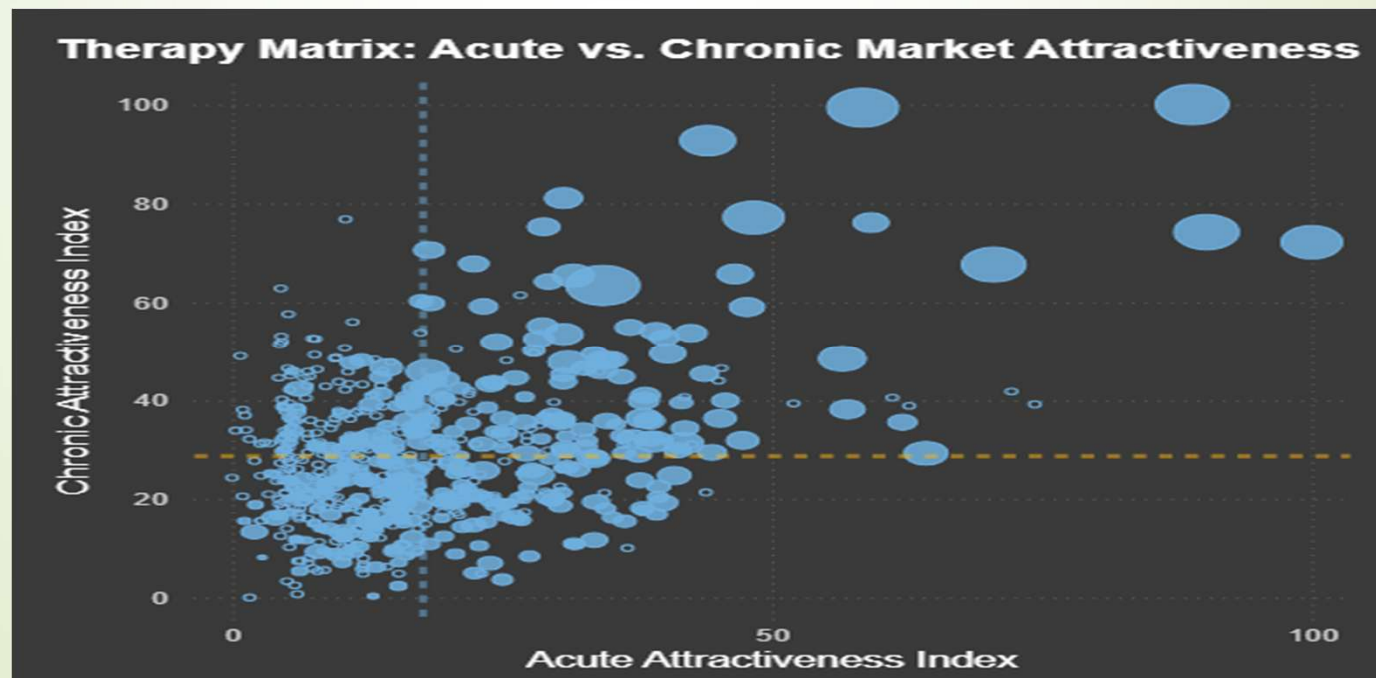
Visual Insights: Geographic Concentration

- ▶ Format: Geographic Bubble Heatmap across the Indian subcontinent.
- ▶ Executive Value: Identifies spatial clusters and supply chain hubs.
- ▶ Action: Optimizes logistics and regional warehousing based on high-density MAI scores.



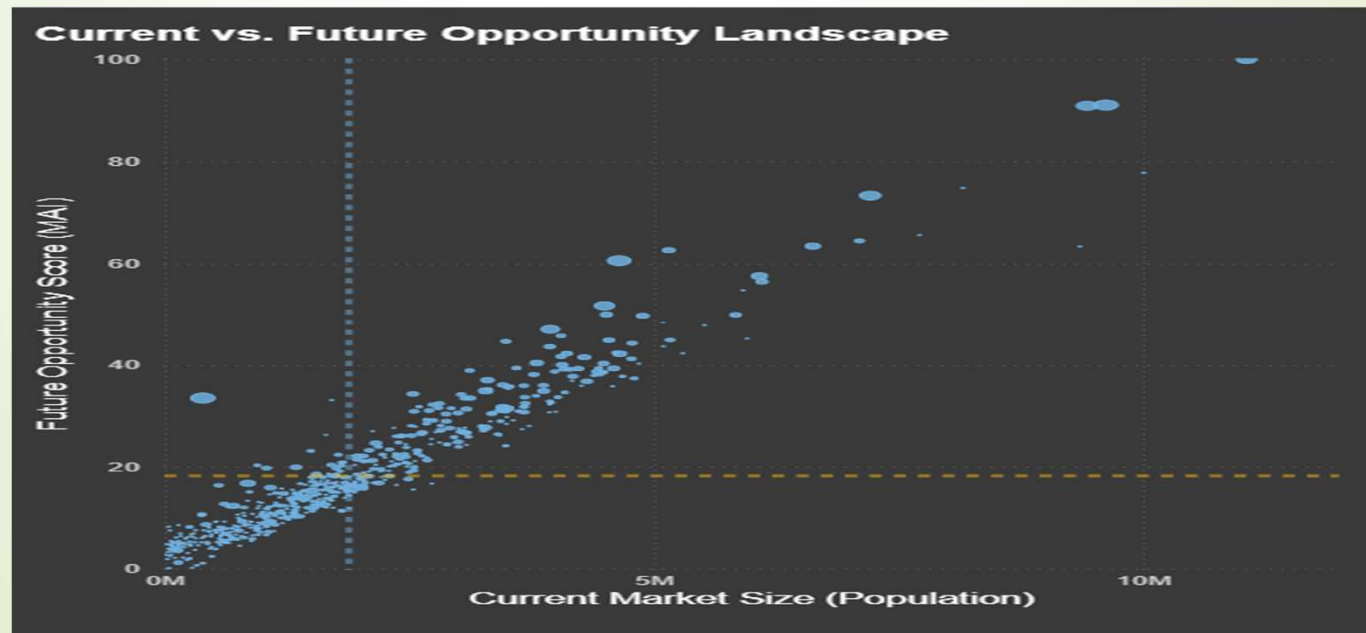
Visual Insights: The Therapy Matrix

- Format: Scatter Plot comparing Acute (X-axis) vs. Chronic (Y-axis) MAI scores.
- Executive Value: Creates a four-quadrant matrix classifying district typologies.
- Action: Identifies 'Gold Mine' districts scoring high in both, vs. niche single-therapy markets.



Visual Insights: Current vs. Future Opportunity

- Format: BCG-style Matrix (Current Population vs. MAI Score).
- Executive Value: Separates mature 'Cash Cow' markets from emerging 'Star' markets.
- Action: Balances the portfolio between harvesting current revenue and investing in future growth.



Strategy: Market Prioritization & Territory Planning

- Redefines territories based on data-driven potential rather than geographical convenience.
- Establishes strict KPIs based on a district's specific MAI baseline.
- Eliminates bias in sales targets.

Strategy: Sales Force Deployment

- High Chronic MAI: Deploy Key Account Managers (KAMs) for specialist doctors.
- High Acute MAI: Deploy mass-market MRs for broad pharmacy coverage.
- Low MAI / High Pop: Rely on wholesale channel distribution.

Replicability & Scalability

- Cross-Therapy: Variables can be swapped (e.g., UV exposure for Dermatology).
- Cross-Geography: Python/Power Query architecture applies to any global market data.
- Dynamic Weighting: The algorithm adapts automatically to new data inputs.

Conclusion & Next Steps

- The MAI transitions Sun Pharma from historical reporting to predictive strategy.
- Immediate Next Step: Integrate final Power BI dashboard with CRM deployment systems.
- Long-Term Vision: Continuous ML training to refine predictive accuracy year-over-year.